LSTM Versus Transformer for Symbolic Music

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***Abstract*—Symbolic music refers to the digital or computational representation of music using explicit data rather than actual sound waves or audio waveforms. Instructions are given for how to reproduce the melody—such as tempo, note onset, offset, duration, and pitch. This allows methods traditionally used for language modeling to also be applied to modeling the sonic art. The need for an objective evaluation of the various methods for achieving this goal motivates the present study. We compared two models—a recurrent neural network with Long Short-Term Memory (LSTM) and a decoder-only Transformer neural network. Each architecture was trained with two different tokenizers (tools that convert musical events into numerical sequences): a custom event-based tokenizer, and REMI+ via MidiTok (an open-source library for MIDI tokenization). Their common objective is to predict the next token. To ensure a fair comparison, the two architectures have a comparable number of parameters (4,087,316 vs. 3,817,984) and were trained on the same data with the same data split. Quality was evaluated through cross-entropy loss, perplexity, Top-1/Top-5 accuracy, per-positional-interval loss, and bits per second of musical time. For both tokenizations, the Transformer achieved lower perplexity than the LSTM, and its advantage grew as the available context increased. The results confirm the hypothesis that the Transformer copes better with long-term dependencies in musical sequences.**

***Index Terms*—LSTM, MIDI, music generation, REMI+, symbolic music, Transformer.**

I. Introduction

Automatic modeling of symbolic music treats musical information as a structured sequence of events rather than as an audio signal. The MIDI format allows the representation of pitch, duration, tempo, instrument, and other musical parameters, which makes the task suitable for methods close to language modeling. In this context, musical events are encoded as tokens—numerical representations of individual MIDI instructions—and the model predicts the next token by analogy with predicting the next word in text. The final token sequence is decoded back into a MIDI file.

Unlike natural language, polyphonic music contains simultaneous events. The model must account both for the horizontal structure of the music—for example, the development of melodic motifs over time—and for the vertical structure—for example, chords and simultaneously sounding notes. This makes the choice of architecture and tokenization especially important: one architecture may be better suited to long dependencies, while one tokenization may provide a clearer metric context (the bar/beat grid—time signatures, bar lines, and beat positions) or a more compact representation of musical time.

The present study compares two types of neural architectures: a recurrent Long Short-Term Memory (LSTM) model [1] and a decoder-only Transformer [2], trained on two representations of MIDI music: a custom event-based tokenizer and REMI+ tokenization via MidiTok [3]. The hypothesis is that the Transformer's advantage will grow as the available context increases. The results confirmed this hypothesis and provide a basis for a concrete recommendation for choosing an architecture—of practical interest to researchers and developers of automatic music-generation systems.

The object of the study is the automatic modeling of polyphonic MIDI sequences. The subject of the study is the influence of architecture and tokenization on the accuracy of predicting the next musical token.

II. Related Work

Recurrent neural networks, including LSTM, are a traditional choice for sequential tasks because they maintain a hidden state that is updated as each token is processed. The LSTM architecture was proposed by Hochreiter and Schmidhuber [1] as a solution to the problems of long temporal dependencies and vanishing gradients. For very long sequences, however, information from the early steps may gradually weaken due to the limited capacity of the hidden state.

The Transformer architecture, introduced by Vaswani et al. [2], offers a self-attention mechanism through which each token can directly use information from preceding tokens in the context window. This makes it a natural candidate for comparison on musical sequences, in which motifs, rhythmic patterns, and harmonic dependencies can appear across a larger number of tokens.

In symbolic music modeling, tokenization plays a role similar to that of the vocabulary in language models. Huang and Yang [4] introduced the REMI representation, structured around bars and positions within the bar, with the aim of more clearly capturing the metric context. Event-based tokenizers typically serialize music as a chronological sequence of MIDI instructions with a more compact token density. The MidiTok library [3] standardizes various tokenization schemes, including REMI+, and provides tools for encoding and decoding MIDI files.

III. Aim, Research Questions, and Hypothesis

The aim of the study is to compare LSTM and Transformer on a next-token prediction task over MIDI sequences and to assess how the choice between the event-based tokenizer and

TABLE I
DATASET SPLIT

Set	Criterion	Share	Count
Train	First 80% of performer groups	∼80%	29,645
Validation	Next 10% of groups	∼10%	4,670
Test	Last 10% of groups	∼10%	3,728

REMI+ affects the results, with the goal of formulating a recommendation for choosing an architecture.

The study is organized around the following questions:

1) Does the Transformer yield lower error and higher Top-5 accuracy than the LSTM under the same tokenization?
2) Does the Transformer's advantage increase in the later positional intervals of the context window?
3) How does the interpretation of the results change when using the event-based tokenizer versus REMI+?
4) Can bits per second of musical time provide a fairer comparison between tokenizations with different vocabularies and different token densities?

The main hypothesis concerns the architecture axis and predicts the answers to questions 1 and 2: the Transformer will achieve lower perplexity and higher Top-5 accuracy than the LSTM within the same tokenization, with its advantage being more visible in the later positional intervals. This hypothesis is tested through the per-positional-interval loss. Questions 3 and 4 are exploratory — they concern the tokenizer axis and the cross-tokenizer metric — and carry no directional prediction.

IV. Materials and Methods

A. Data

The experiments use the Lyrics-MIDI Dataset from HuggingFace [5]. The full set contains approximately 179,000 MIDI files, and after deduplication—38,043 files. The genre distribution is dominated by rock, metal, and pop. The files follow a naming convention of the form <song_name> --- <band_name> --- <hash>.mid, which allows the performer's name to be extracted directly from the file name.

The data split is done by performer, not by individual song. All songs of one performer fall into the same set. A random split by individual song would allow the model to be tested on pieces by performers present in the training set. The models could "memorize" the style of a given performer from the training data, which would artificially inflate test-set accuracy without reflecting the real ability to generalize. Splitting by performer is a formal way to limit data leakage between the training, validation, and test sets. The groups are sorted before shuffling with a fixed seed (42), after which they are distributed in an approximate ratio of 80% training, 10% validation, and 10% test (Table I).

B. Tokenizations

The first tokenization is a custom event-based implementation developed for this project. It serializes the merged MIDI tracks into a single one-dimensional, chronologically ordered stream of tokens. Flattening into one stream does not discard polyphony: simultaneous notes (chords) appear as consecutive tokens separated by a zero time-shift, so the vertical structure is preserved within the linear sequence and the model must reconstruct both the vertical (simultaneous) and horizontal (temporal) structure from it. The vocabulary contains 532 tokens: NOTE_ON, NOTE_OFF, DRUM_ON, TIME_SHIFT, VELOCITY, PROGRAM, TEMPO, BAR, PAD (index 529), BOS (530), and EOS (531). Unwanted MIDI events such as Sustain Pedal, Pitch Bend, and SysEx are ignored during encoding. PAD is used only to pad windows and is ignored by the loss function.

TABLE II
COMPARISON OF THE TWO TOKENIZATIONS

Characteristic	Event-based	REMI+
Core idea	Chronological MIDI events	Bars and positions
Vocabulary	532 tokens	502 tokens
Time	TIME_SHIFT, 10 ms	Duration and Position
Metric context	BAR and TEMPO; no TimeSig	Bar, Position, Tempo, TimeSig
Token density	Lower	Higher
Interpretation	More compact chronology	More explicit musical structure

The second tokenization is REMI+ via the MidiTok library, version 3.0.6.post1 [6]. It represents music through Bar, Position, Pitch, Velocity, Duration, Tempo, Program, and TimeSig tokens. The vocabulary size is 502 tokens. The advantage of REMI+ is its more explicit structuring around bars and positions within the bar, but with a fixed context of 512 tokens it may cover less real musical time due to a higher token density for the same musical content. Table II summarizes the two schemes.

C. Model Architectures

Two architectures are compared. The LSTM model has two recurrent layers, a hidden dimension of 512 neurons, and an embedding dimension of 256. Each step takes the current token, updates the hidden state through input, forget, and output gates, and predicts the next token through a linear projection onto the vocabulary.

The Transformer is a decoder-only architecture with a causal mask. The configuration includes 4 layers, 4 self-attention heads, an embedding dimension of 256, and a Feed-Forward Network (FFN) dimension of 1280. The causal mask guarantees that, when predicting position i, the model has access only to positions 0 through $i-1$. The weights of the embedding layer and of the output projection are shared (weight tying), which reduces the number of parameters. The FFN dimension of 1280 was chosen so that the total number of trainable parameters falls within $\pm 10\%$ relative to the LSTM. Table III lists both configurations and the controlled parameters.

D. Experimental Combinations

To separate the effect of the architecture from the effect of the tokenization, four combinations are trained (Table IV).

TABLE III
ARCHITECTURES AND CONTROLLED PARAMETERS

Component	LSTM	Transformer
Embedding	256	256
Structure	2 LSTM layers	4 decoder-only layers
Hidden / FFN	512	1280
Attention heads	—	4
Parameters	4,087,316	3,817,984
Ratio	1.000	0.934 (±10%)
Context	512 tokens	512 tokens

TABLE IV
THE FOUR EXPERIMENTAL COMBINATIONS

Comb.	Architecture	Tokenization	Role
A	LSTM	Event-based	Baseline
B	Transformer	Event-based	Comparison with A
C	LSTM	REMI+	Comparison with A
D	Transformer	REMI+	Comparison with C and B

TABLE V
DISTRIBUTION OF TRAINING ACROSS PLATFORMS

Combination	Platform
A	Kaggle (NVIDIA Tesla T4)
B	Kaggle (NVIDIA Tesla T4)
C	Lightning AI (NVIDIA A10G)
D	Lightning AI (NVIDIA A10G)

Within a single tokenization, the architectures can be compared directly through cross-entropy loss, perplexity, Top-k accuracy, and per-positional-interval loss. Across different tokenizations these metrics are not sufficient due to the different vocabulary size and different token density, so normalization through bits per second of musical time is used.

E. Preparation and Training

Data preparation is performed once. After the group-based split, pre-tokenization is carried out separately for each tokenization. The resulting `.pt` files are loaded on every training run without re-tokenization. The data component extracts windows of length 512 tokens via a non-overlapping stride of 512 positions, where the target sequence is the input sequence shifted one position forward. The last incomplete window is padded with PAD, and PAD is ignored when computing the loss.

Training uses the Adam optimizer with a learning rate of 3×10^{-4}, linear warmup over the first 1000 steps, cosine decay to 10% of the initial value, and categorical cross-entropy loss ignoring PAD. Gradients are clipped to a maximum norm of 1.0. On each new minimum of the validation loss, a checkpoint is saved with the model weights, the optimizer state, and the current epoch. The four combinations were trained across two cloud platforms (Table V).

F. Metrics

Within a single tokenization, the main metrics are cross-entropy loss, perplexity, Top-1 accuracy, Top-5 accuracy, and per-positional-interval loss. The cross-entropy loss measures the average negative log-probability of the correct token:

$$\mathrm{CE} = -\frac{1}{N}\sum_{t}\log p_\theta(y_t \mid x_{\le t}) \tag{1}$$

where N is the number of evaluated tokens, and $p_\theta(y_t \mid x_{\le t})$ is the probability assigned by the model to the correct token at position t.

$$\mathrm{PPL} = \exp(\mathrm{CE}) \tag{2}$$

Top-1 accuracy indicates whether the most probable predicted token is correct; Top-5 accuracy checks whether the correct token is among the five most probable. The latter is especially useful for music, because more than one acceptable continuation exists.

The per-positional-interval loss groups the loss by position in the context into four intervals: 0–127, 128–255, 256–383, and 384–511. If the Transformer copes better with longer dependencies, the difference between it and the LSTM should grow in the later intervals.

For comparison across different tokenizations, bits per second of musical time is used:

$$\mathrm{Bits/s} = -\frac{1}{T_{\mathrm{sec}}}\sum_{t}\log_2 p_\theta(y_t \mid x_{\le t}) \tag{3}$$

where T_{sec} is the real musical duration of the input sequence in seconds, computed by summing the time values in the token sequence. This normalization reduces the effect of the different vocabulary size and the different number of tokens for the same real musical content.

V. EXPERIMENTAL DESIGN

The experimental design aims for a fair comparison by controlling the main variables. All models use the same splits, the same context length, the same optimizer with an identical configuration, the same principle for saving the best checkpoint, and a comparable number of trainable parameters.

The main independent variable is the model architecture, and a secondary independent variable is the tokenization. The dependent variables are the prediction-quality metrics. The control variables include context length, splits, preprocessing, tokenizer configurations, training loop, and loss function. Table VI lists the fixed parameters used for reproducibility.

TABLE VI
CONTROL PARAMETERS FOR REPRODUCIBILITY

Parameter	Value
Seed	42
PyTorch	≥2.0 (cu124 for local evaluation)
Context	512 tokens
Stride	512 (no overlap)
Batch size	32
Learning rate	3×10^{-4}
Warmup	1,000 steps
Max. epochs	30
Gradient clipping	max_norm = 1.0

TABLE VII
MAIN METRICS BY EXPERIMENTAL COMBINATION

C.	Tok.	Model	CE ↓	PPL ↓	Top-1 ↑	Top-5 ↑
A	Event	LSTM	1.2901	3.63	61.47%	88.22%
B	Event	Transf.	1.1678	3.21	66.38%	89.33%
C	REMI+	LSTM	0.9299	2.53	71.97%	92.57%
D	REMI+	Transf.	0.7017	2.02	79.31%	94.60%

TABLE VIII
PER-POSITIONAL-INTERVAL LOSS

Comb.	0–127 ↓	128–255 ↓	256–383 ↓	384–511 ↓
A: LSTM/Event	1.5828	1.2179	1.1821	1.1709
B: Tr./Event	1.5590	1.1161	1.0194	0.9669
C: LSTM/REMI+	1.1849	0.8655	0.8367	0.8280
D: Tr./REMI+	1.0703	0.6441	0.5638	0.5216

VI. RESULTS

A. Summary Results Within a Single Tokenization

In the comparison within the event-based tokenizer (A vs. B), the Transformer achieves a lower cross-entropy loss by 0.1223 units (1.1678 vs. 1.2901), a lower perplexity by 0.42 (3.21 vs. 3.63), and a higher Top-1 accuracy by 4.91 percentage points (66.38% vs. 61.47%) (Table VII). For the REMI+ tokenization (C vs. D), the differences are more pronounced: the CE loss is lower by 0.2282 units (0.7017 vs. 0.9299), perplexity is lower by 0.51 (2.02 vs. 2.53), and Top-1 accuracy is higher by 7.34 percentage points (79.31% vs. 71.97%). The Transformer surpasses the LSTM on all metrics within both tokenizations.

B. Per-Positional-Interval Loss

The difference in cross-entropy loss between the LSTM and the Transformer grows as the position in the context window advances (Table VIII). For the event-based tokenizer, the difference (A − B) is 0.024 in the interval 0–127, 0.102 in 128–255, 0.163 in 256–383, and 0.204 in 384–511—an increase of 8.5× from the first to the last interval. For REMI+, the difference (C − D) is 0.115, 0.221, 0.273, and 0.306, respectively. The Transformer's advantage is systematically larger with longer available context, which is the main evidence in support of the hypothesis.

C. Comparison Across Tokenizations

In the comparison via bits per second of musical time, REMI+ provides lower uncertainty for both architectures (Table IX). For the LSTM, the difference is 39.18 bits/s in favor of REMI+ (94.49 vs. 133.67). For the Transformer, the difference is larger—49.70 bits/s (71.30 vs. 121.00). The Transformer with REMI+ achieves the lowest value of all four combinations (71.30 bits/s), which means the greatest predictive confidence per unit of musical content.

D. Qualitative Assessment of Generation

In addition to the quantitative metrics, an informal qualitative assessment of the models' generations was performed. It was carried out independently by both authors; we stress that it is an illustrative evaluation, not a controlled listening study with a panel of external raters, and that it concerns the overall quality of generation rather than the position-dependent advantage analyzed in Section VI-B. Five input prompts were deliberately selected to span the structural range of symbolic music—from very sparse, slow, melodic material to dense, fast, percussive material—so that the assessment would not be biased toward a single texture (Table X). Each prompt was truncated to its first 256 tokens—half of the model's 512-token context window, leaving room for the continuation—and continued for 4096 tokens by each of the four combinations (A–D) under three decoding settings: greedy decoding (T=0, always taking the highest-probability token) and top-k sampling (k=40, drawing the next token from the 40 highest-probability candidates after temperature scaling) at T=0.5 and T=1.0.

TABLE IX
METRICS FOR COMPARISON ACROSS TOKENIZATIONS

Comb.	Tokenizer	Model	bits/s ↓
A	Event-based	LSTM	133.67
B	Event-based	Transformer	121.00
C	REMI+	LSTM	94.49
D	REMI+	Transformer	71.30

TABLE X
STRUCTURAL SPREAD OF THE INPUT PROMPTS (MEASURED ON THE SOURCE MIDI)

Prompt	Genre	Tempo	Density (notes/s)	Pitch range
Casta Diva	classical / opera	74	0.7	strings + voice
Blue Jean Blues	blues	46	1.6	guitar
'Till I Collapse	rap	90	3.5	wide
Aerials	nu-metal	203	5.2	∼2 oct.
Tornado of Souls	thrash	195	10.3	∼6 oct.

TABLE XI
MEAN OBJECTIVE DESCRIPTORS BY DECODING SETTING (20 CLIPS EACH; PITCH-CLASS ENTROPY IN BITS, MAX $\log_2 12 = 3.58$; DOMINANT SHARE = FRACTION OF NOTES EQUAL TO THE SINGLE MOST-COMMON PITCH)

Decoding	Distinct pitches	Pitch-class entropy	Dominant-pitch share
Greedy (T=0)	13.2	1.42	0.47
T=0.5	15.9	1.68	0.38
T=1.0	38.9	2.94	0.13

1) Decoding strategy is the main driver of listenability: Greedy decoding and T=0.5 collapse for *every* combination into repeated single notes or chords; only T=1.0 recovers musical variety. Table XI quantifies this (each row averages the 20 continuations produced under that setting): the mean number of distinct pitches rises from 13.2 (greedy) and 15.9 (T=0.5) to 38.9 (T=1.0), mean pitch-class entropy from 1.42 to 1.68 to 2.94 bits, and the dominant-pitch share falls from 0.47 to 0.13. The transition is not gradual—T=0.5 patterns with greedy decoding, not with T=1.0.

TABLE XII
LISTENABILITY AT T=1.0 (0 = UNLISTENABLE, 1 = PARTIAL, 2 = LISTENABLE, 3 = MOST NATURAL)

Prompt	A lstm/ev	B tf/ev	C lstm/remi	D tf/remi
Casta Diva	1	1	2	3
Blue Jean Blues	1	0	2	1
'Till I Collapse	2	2	2	1
Aerials	2	0	2	3
Tornado of Souls	3	1	1	3

2) Tokenization and architecture at T=1.0: Both REMI+ combinations (C, D) produced listenable continuations on three to four of the five prompts, whereas the event-tokenizer combinations were markedly less stable—ranging from coherent (LSTM+event on the dense thrash prompt) to a degenerate, near-silent collapse (Transformer+event on the sparse blues prompt: 0.02 notes/s). Within the listenable tier, Transformer+REMI+ (D) produced the most musically natural continuations, judged "flows fluently / sounds good / most natural" on three of the five prompts—consistent with its lowest cross-entropy (0.70). LSTM+REMI+ (C) was less striking but the most reliable (Table XII). The same trend is visible objectively: at T=1.0 the REMI+ continuations average higher pitch-class entropy (3.06 vs. 2.82 bits) and a lower dominant-pitch share (0.10 vs. 0.16) than the event-tokenizer continuations. This audible ranking corroborates the quantitative results.

VII. DISCUSSION

We first answer the four research questions of Section III directly:

1) *Does the Transformer yield lower error and higher Top-5 accuracy under the same tokenization?* Yes. For both tokenizations it attains lower cross-entropy and higher Top-5 accuracy than the LSTM (Table VII: Top-5 rises from 88.22% to 89.33% for the event tokenizer and from 92.57% to 94.60% for REMI+).
2) *Does its advantage increase in the later positional intervals?* Yes. The cross-entropy gap grows monotonically across the four intervals (Table VIII), by 8.5× from the first to the last interval for the event tokenizer (0.024 → 0.204) and from 0.115 to 0.306 for REMI+.
3) *How does the interpretation change between the event tokenizer and REMI+?* The Transformer's advantage is larger under REMI+ (absolute CE gap 0.2282 vs. 0.1223), indicating that the more explicit metric structure of REMI+ benefits self-attention more than it benefits the LSTM.
4) *Can bits per second of musical time provide a fairer cross-tokenization comparison?* Yes. It normalizes for the differing vocabulary size and token density and yields a consistent ordering (Table IX), under which REMI+ is favorable for both architectures and Transformer+REMI+ is best overall.

The results confirm the main hypothesis: the Transformer surpasses the LSTM on all quantitative metrics within both tokenizations. The growing per-positional-interval gap (Table VIII) is in direct agreement with the self-attention mechanism: the Transformer has direct access to all preceding tokens, whereas the LSTM passes context through a fixed-size hidden-state vector whose capacity saturates for longer dependencies.

The Transformer's advantage is also larger for REMI+ than for the event-based tokenizer: its clearer metric structure—explicit Bar, Position, and TimeSig tokens—provides a richer context for the self-attention mechanism, which the Transformer exploits more effectively than the LSTM.

In the comparison across tokenizations via bits per second of musical time, REMI+ yields lower uncertainty for both architectures. Despite the higher token density, the more structured representation compensates for the need for more tokens for the same musical content. The larger difference for the Transformer (49.70 bits/s versus 39.18 bits/s for the LSTM) suggests that the Transformer benefits disproportionately from the richer REMI+ representation.

The qualitative assessment (Section VI-D) is broadly consistent with the quantitative ranking: the combination with the lowest cross-entropy, Transformer+REMI+, also produced the continuations judged most natural. We caution, however, against reading the listening assessment as independent confirmation that the Transformer surpasses the LSTM in *perceived* quality. The agreement is clean only under REMI+; under the event tokenizer the architectural effect on listenability is mixed—Transformer+event was the least stable combination and on one prompt collapsed to near-silence—and the most reliable combination by ear was LSTM+REMI+, not the Transformer. As an informal, single-setting check, the assessment corroborates the metric-based ordering of the best system but does not, on its own, establish an architectural advantage in perceived musical quality.

The results are consistent with the findings of Huang and Yang [4], who show that a Transformer with bar-based tokenization (REMI) generates musically more coherent sequences compared to LSTM-based approaches for piano music. Our findings confirm this trend on a more diverse dataset dominated by rock and pop genres.

A distinctive feature of the present study is that the Transformer's advantage is measured quantitatively per positional interval—an approach that Huang and Yang do not apply as such a positional breakdown, yet it is precisely this that allows us to localize exactly when and by how much the difference between the two architectures grows as context increases.

Vaswani et al. [2] demonstrate the advantage of self-attention over recurrent models on language-modeling tasks, motivating it by the direct access to long dependencies without information decay. Our results on musical sequences are analogous and extend the observation to a new application domain—symbolic modeling of polyphonic music. The growing per-positional-interval difference provides concrete empirical evidence that the self-attention mechanism retains its advantage for long dependencies even outside the domain of natural language.

What distinguishes the present study from related work is the dual controlled design: architecture and tokenization are treated as independent factors, and the comparison between tokenizations of different density and different vocabulary size is carried out via bits per second of musical time—a metric that is not standard practice in related studies. It enables a fair comparison where a direct comparison of cross-entropy loss would be misleading.

VIII. Limitations and Threats to Validity

The study has several limitations. First, the context of 512 tokens covers approximately 30 seconds of music for the event-based tokenizer, and less for REMI+ due to its higher token density. This limits the models' ability to learn structure at the level of verse, chorus, and overall song form. A more in-depth study of the Transformer's advantages for long dependencies would require contexts of 1024 tokens or more.

Second, the event-based tokenizer uses 10 ms quantization, which introduces small temporal inaccuracies in fast musical passages. When decoding percussion instruments, events are assigned a fixed short duration, which does not capture all the nuances of percussive performance.

Third, training was distributed between Kaggle and Lightning AI. Despite the identical code configuration and the fixed seed, the different GPU types and library versions may introduce small variations in the stochastic optimization process.

Fourth, the subjective listening to the generated MIDI examples is only an informal qualitative check and is not a statistically valid measure.

IX. Conclusion

The paper presents a recommendation for choosing a neural architecture for a next-musical-token prediction task, supported by a controlled experimental comparison of LSTM and Transformer with two different tokenizers and a comparable number of parameters. The main contribution is the clear separation of two factors—architecture and tokenization—through a set of metrics, including per-positional-interval loss and bits per second of musical time.

The results confirmed the hypothesis: the Transformer surpasses the LSTM on all metrics for both tokenizations. The key scientific finding is that the LSTM–Transformer cross-entropy gap grows $8.5\times$ across the context window for the event-based tokenizer—concrete empirical evidence that self-attention retains and amplifies its advantage exactly where the LSTM struggles with long dependencies. REMI+ provides lower uncertainty in bits per second for both architectures, and the Transformer benefits from the richer REMI+ representation to a greater degree than the LSTM.

For science, the present study contributes two elements that are not standard in related publications: a positional breakdown of the loss as a tool for localizing the architectural advantage, and the bits-per-second-of-musical-time metric as a fair way to compare tokenizations of different density.

TABLE XIII
Detailed Configuration of the LSTM Model

Component	Parameters
Embedding (532×256)	136,192
LSTM layer 1 (input 256, hidden 512)	1,576,960
LSTM layer 2 (input 512, hidden 512)	2,101,248
Output projection (512×532 + bias)	272,916
Total	4,087,316

TABLE XIV
Detailed Configuration of the Transformer

Component	Value
Embedding dimension	256
Decoder-only layers	4
Self-attention heads	4
FFN dimension	1280
Weight tying	Yes
Total trainable parameters	3,817,984

For practice, the recommendation is unambiguous: under a limited computational budget, the Transformer with REMI+ tokenization achieves the best quality for symbolic music modeling with a context of 512 tokens.

Future work of interest would include: extending the context window; including additional architectures; conducting a formal listening study with multiple external raters; and investigating whether the observed effects hold for progressive, classical, or jazz music.

Appendix: Model Parameters

Table XIII gives the per-component parameter breakdown of the LSTM model, and Table XIV the configuration of the Transformer.

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